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A WIRELESS NETWORK USER LOCATION STUDY

Methodological instructions for laboratory work

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1. A WIRELESS NETWORK USER LOCATION STUDY

1.1. Work objective

Conduct a survey on the use of location information.

1.2. Theoretical material

Location information can be used to provide new value-added services. In this paper, we will study the location information of a wireless local area network (WLAN) user. In order to use location information, it is necessary to have a certain infrastructure. You can have your own infrastructure or use a location service provider's network. We will use pre-prepared infrastructure in our work.

Its structure is presented in Fig. 1

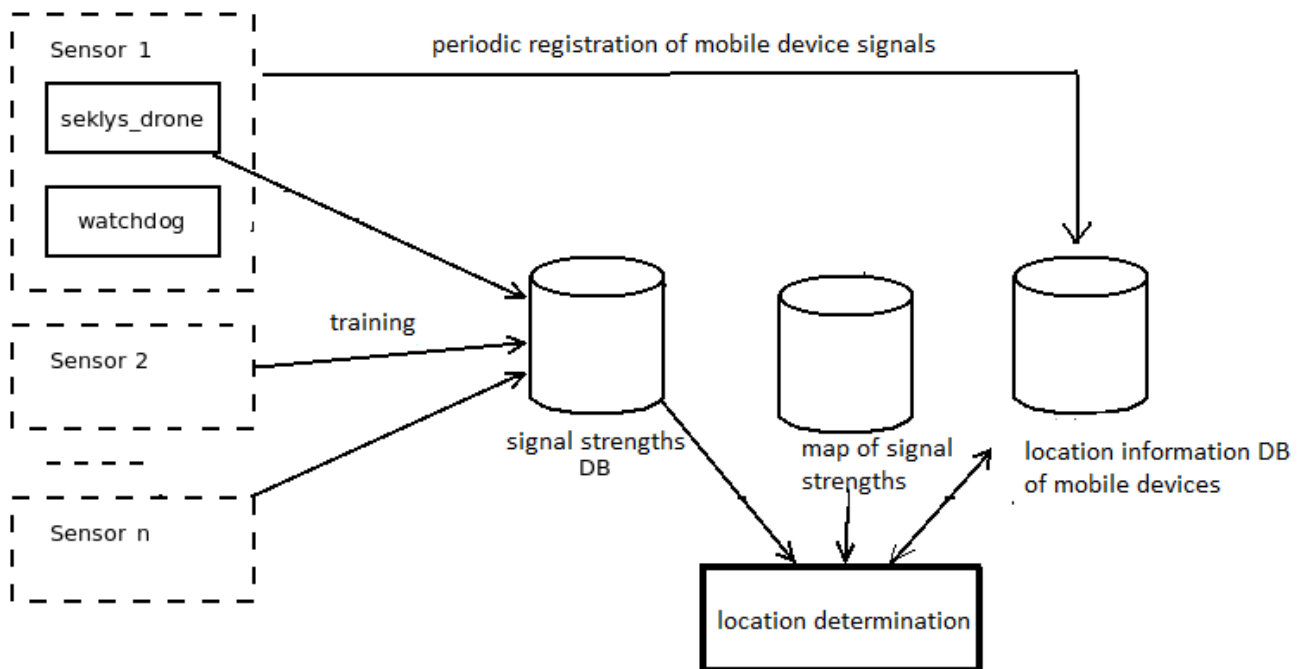


Fig. 1 . A wireless network user location system

Each user using WLAN network communicates with the Access Points (AP - Access Point), and the AP "see" each user. They receive various information about the user, including the strength of the signal sent by the user. The further the user is from the AP, the weaker the signal reaches the AP. With a wireless network infrastructure that includes at least three stations (AP) capable of recording the signal strength of wireless network users, it is possible to determine the location of a wireless network user.

For this purpose, the space to be located is divided into a grid of a certain size. The system is trained by placing a test user in each grid cell and recording the strength of the signal it sends. The grid and locations where the signal strength was measured are presented in Fig. 2.

It shows the 4th floor plan of the Electronics building. "1" corresponds to the cell where signal strengths were recorded during "training".

The "visible" signal strength of at least 3 AP stations is recorded in each cell.

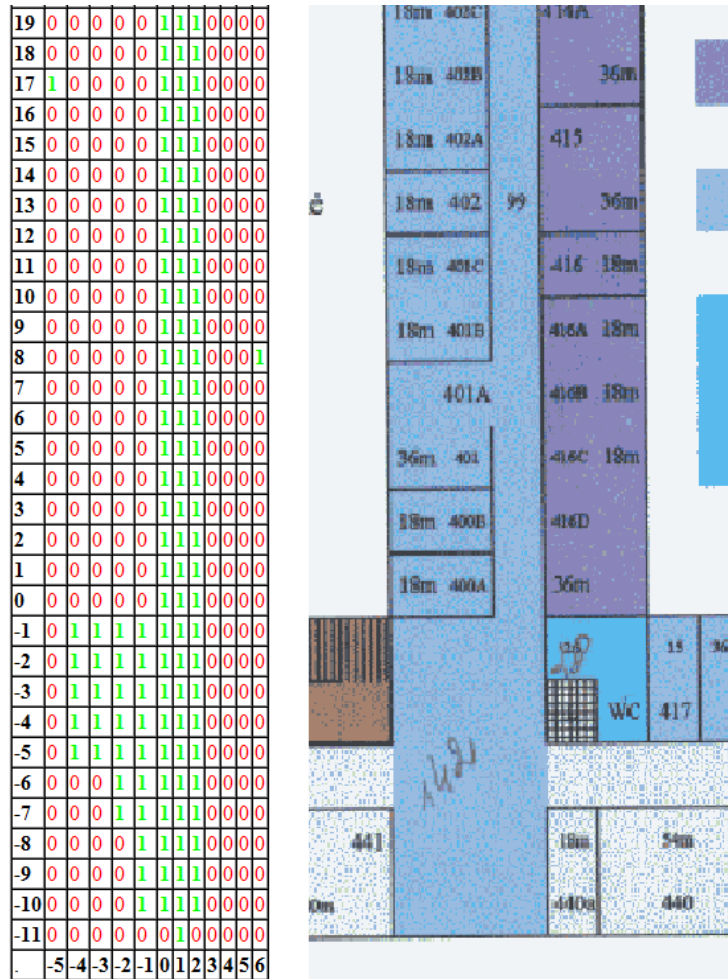


Fig. 2 Signal strength measurement points

The map of signal strengths created in this way can be used to find the location of other users. The search is carried out using the "Nearest Neighbor" search method [1,2].

The data for the search algorithm (map of signal strengths) is stored in a MySQL database. You can connect to it remotely and get the necessary information.

The data is presented in the following table:

No.	Title	Value
1.	Location server	sekllys.ila.lt
2.	phpMyAdmin	https://sekllys.ila.lt:5443/phpmyadmin

3.	MySQL database parameters: Database Username Password	LDB (in capital letters) stud vLXCDmSG6EpEnhXX
4.	Table of signal strengths	stiprumai
5.	Table of measurements	matavimai
6.	Location information table for mobile devices	vartotojai

The structure of the "stiprumai " table:

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
matavimas	int(11)	NO		NULL	
sensorius	varchar(10)	NO		NULL	
stiprumas	int(4)	NO		NULL	

The structure of the „matavimai“ table:

Field	Type	Null	Key	Default	Extra
matavimas	int(11)	NO	PRI	NULL	auto_increment
x	int(11)	NO		NULL	
y	int(11)	NO		NULL	
atstumas	float	NO	MUL	NULL	

The structure of the „vartotojai“ table:

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
mac	varchar(17)	NO		NULL	
sensorius	varchar(10)	NO		NULL	
stiprumas	int(4)	NO		NULL	

1.3. Work assignment

Choose the right programming language for creating a mobile application (the one you will be introduced to during the 2nd project) to complete the work tasks.

1. Connect to the location server database.
2. Render a "grid" of measurement points in database tables. It must be clear which grid points have measurements taken and which do not. You should get a picture similar to the one shown in Figure 2. The map shown in the figure is slightly reduced (there are more entries in the table). It shows the rows and columns in the "measurements" table marked with "x" and "y" fields, respectively.
3. Calculate the location of the user (identified by the MAC address) using the nearest neighbor search method. Set the location in the "grid" of the measurement points by specifying the coordinates or highlighting a certain cell with a different color or something. Note that the user table „vartotojai“ may contain several entries about a user from different access points. The station names match the names used in the strength table „stiprumai“.
4. Realize the ability to enter any MAC address with radio signal strengths to the corresponding access station in the system you are building and realize location determination for them.

1.4. Literature and references

1. Description of the nearest neighbor search method (Moodle)
2. Q. Tran, J. Wirawan Tantra, C. Heng Foh, A. Tan, K.Yow, D. Qiu. Wireless Indoor Positioning System with Enhanced Nearest Neighbors in Signal Space Algorithm. Nanyang Technological University, Singapore, Concordia University, Canada, 2006 IEEE. (Moodle)
3. Description of the nearest neighbor search method. http://en.wikipedia.org/wiki/K-nearest_neighbor_algorithm
4. Howto: Connect MySQL server using C program API under Linux or UNIX. <http://www.cyberciti.biz/tips/linux-unix-connect-mysql-c-api-program.html>
5. How to: Connect to my MySQL Database server using command line and php. <http://www.cyberciti.biz/faq/how-to-connect-to-my-mysql-database-server-using-command-line-and-php/>
6. MySQL Commands. <http://www.pantz.org/software/mysql/mysqlcommands.html>